



Traffic Signal Optimization Pilot Proposal

Location: Shenzhen, China

Website: <https://enertraffic.com>

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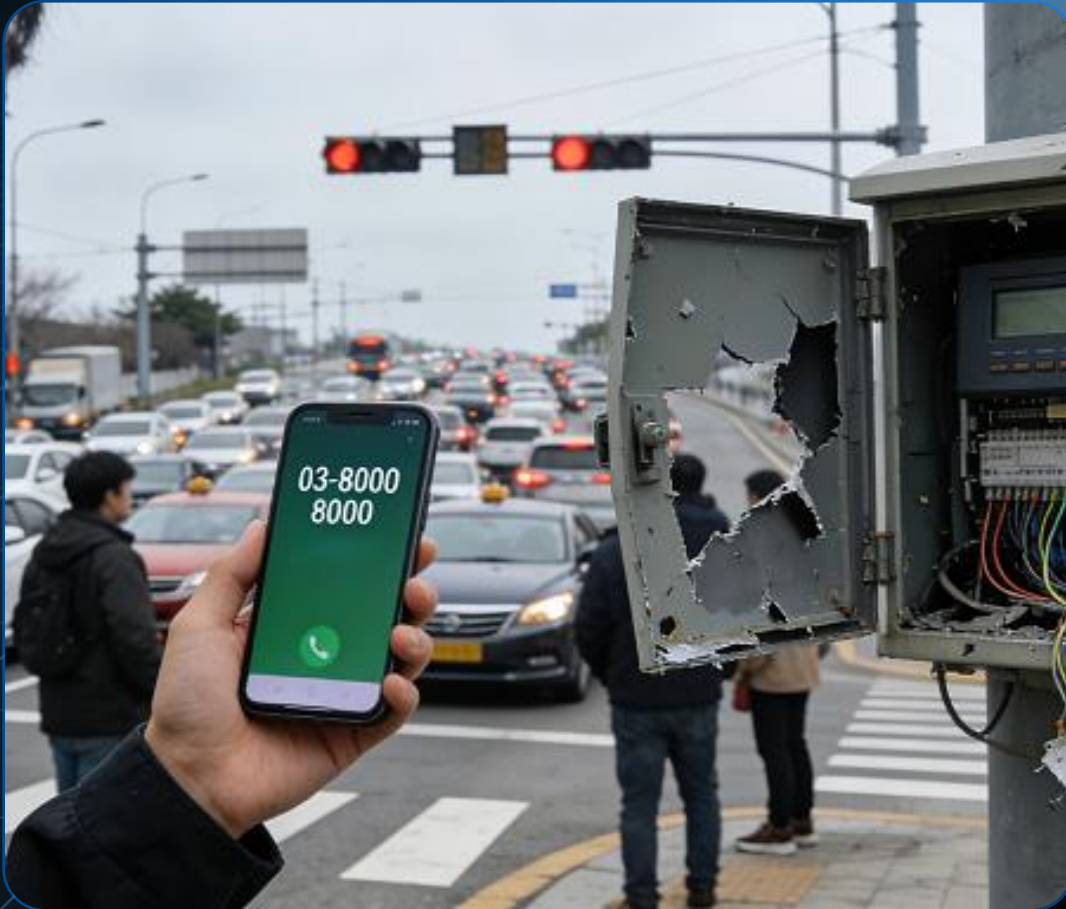


05 Long Term Evolution

01

Challenges of Traffic Management

Challenges in Traffic Signal Management



Unscientific Timing

Relies on experience without data models, leading to resource waste. Fixed timing fails to adapt to dynamic traffic conditions.



Limited Sensing

Traditional sensors offer only simple recognition and limited functionalities. High maintenance costs and long recovery times hinder comprehensive monitoring (e.g. loop traffic detector).



Passive Management

Lack of proactive monitoring means problems are only addressed after complaints, preventing timely improvements.



Isolated Intersections

Lack of coordination between intersections prevents optimal traffic flow distribution.

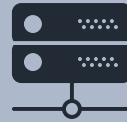
02

Traffic Light Optimization Pilot Solution

Intelligent Transportation Architecture Overview

Unified Management Layer

Signal Control Platform · Data Fusion · Solution Diagnosis Regional Coordination · App Enablement



Unified Signal Control Platform

Precision Control Layer

Built-in Green Wave Coordination & Single-Point Optimization · Real-Time Response · Safety & Stability



Intelligent Signal Controller Edge Computing Hub

Holistic Perception Layer

All-Weather · Full-Element · Flow · Speed · Queue · Trajectory · Event · Vehicle Type



Camera



Radar



Fusion Sensor



RSU



Env. Sensor



Beidou&GPS

Traffic Light Optimiztion Pilot Soution

Solution Principle: Simple and secure, Isolated from the existing network, Easy to interconnect with the existing network, Smooth Scaling and evolution.



Cloud Platform



Data Sensing Layer

Deploy radar-video fusion sensors to collect real-time traffic data to provide accurate information for .precise control of traffic lights.



Precision Control Layer

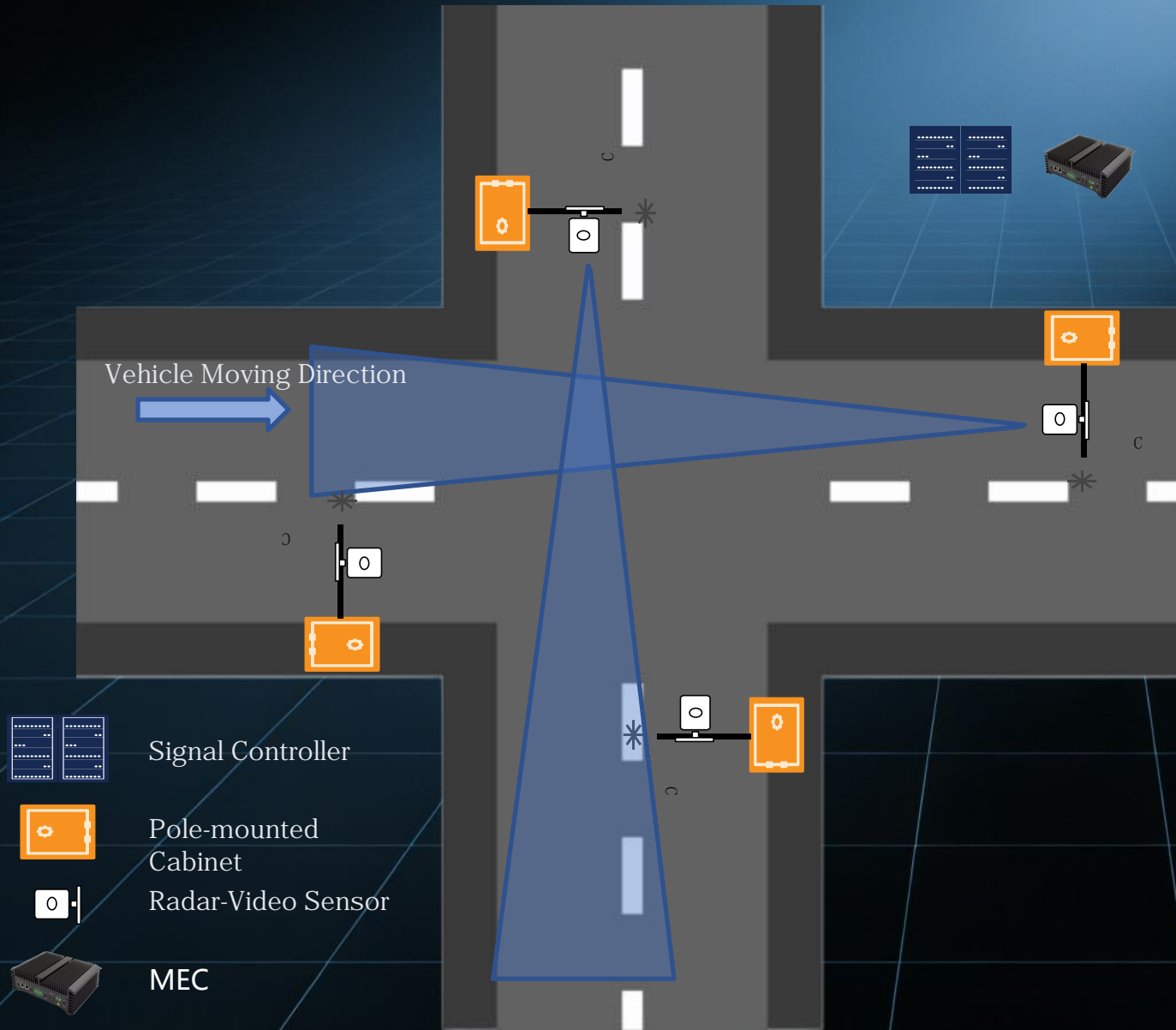
Signal controller + MEC execute optimization strategies, achieving a closed-loop adaptive system from perception to control.



Unified Mng. Layer

Provides unified traffic management system and interconnection with surrounding systems.

Deployment Diagram



Device Type	Qty
Signal controller	1
Radar-video fusion sensor	4
Switch	4
Pole-mounted cabinet	4
MEC	1
Fill light	4 (Optional)

1. A signal controller is required for the entire intersection to control all traffic signal lights at the intersection.
2. 4 radar-video fusion devices are to be installed to face traffic flow at the intersection.
3. An MEC may be configured and installed inside the signal controller cabinet for edge computing.

Core Equipment: Signal Controller



Technical Specifications

- 32 light groups, 96 output terminals & 32 detector inputs
- 32-channel leakage detection & IP65 protection rating
- 12-inch touch screen for easy operation
- Double doors design with manual control & backup power

Key Functions

- Adaptive control
- Multi-period control
- Networked coordination
- Supports manual operation
- Special control modes

Key Advantages of Signal Controller



High Security

Stable operation with low failure rate. Equipped with multiple safety mechanisms including leakage protection, independent yellow flash, configurable degradation strategies etc.



High Compatibility

Supports standard protocols for seamless integration with upper-level platforms. Support 3rd sensors access. Opens private protocols to execute third-party optimization algorithms.



Abundant Functions

Supports rich functions including daily operations, intelligent optimization, and regional coordination.

Core Equipment: Smart Radar



⚙️ *Technical Specifications*

- 📡 Working frequency: 80GHz
- 📏 Detection Range: 350m+
- 🕒 Detectable targets: 200+
- 📷 Resolution: 5MP for traffic detection and LPR, Starlight night vision

🏆 *Key Advantages*

- 🔄 Radar + video fusion technology for enhanced accuracy
- 🌐 Comprehensive Traffic Data Acquisition
- 🕒 Easy Implementation & Maintenance

Core Equipment: MEC

Key Value Proposition

Enabling real-time local data processing and collaborative decision-making at the edge. This ensures low-latency responses and efficient handling of critical traffic data right where it's generated.



Technical Specifications

-  **Processor**
4×Cortex-A76 + 4×Cortex-A55, 2.4GHz
-  **System Memory**
32G DDR5
-  **Operating System**
Linux
-  **Input Voltage**
220V
-  **Interfaces**
USB3.0/HDMI/Ethernet/RS485

Key Advantages of MEC



Flexible Computing Power

With computing power scalable up to 200 TOPS, the MEC can be flexibly configured to meet the demands of simple to highly complex traffic scenarios.



Built-in Core Algorithms

Pre-installed with mature algorithms for video cross-domain fusion, green wave coordination, and single-point optimization etc, significantly shortening deployment cycles.



Rich Interfaces & Expandability

Equipped with USB, HDMI, Ethernet, RS485, and 4G/5G modules, ensuring broad compatibility and easy future expansion.

Single Intersection Optimization Procedure

Single-point Adaptive Optimization

Creates a closed-loop control system from front-end perception, adaptive control, to system evaluation. This improves efficiency and solves issues like uneven green time distribution and wasted green time.

Optimization Process



Front-end Perception

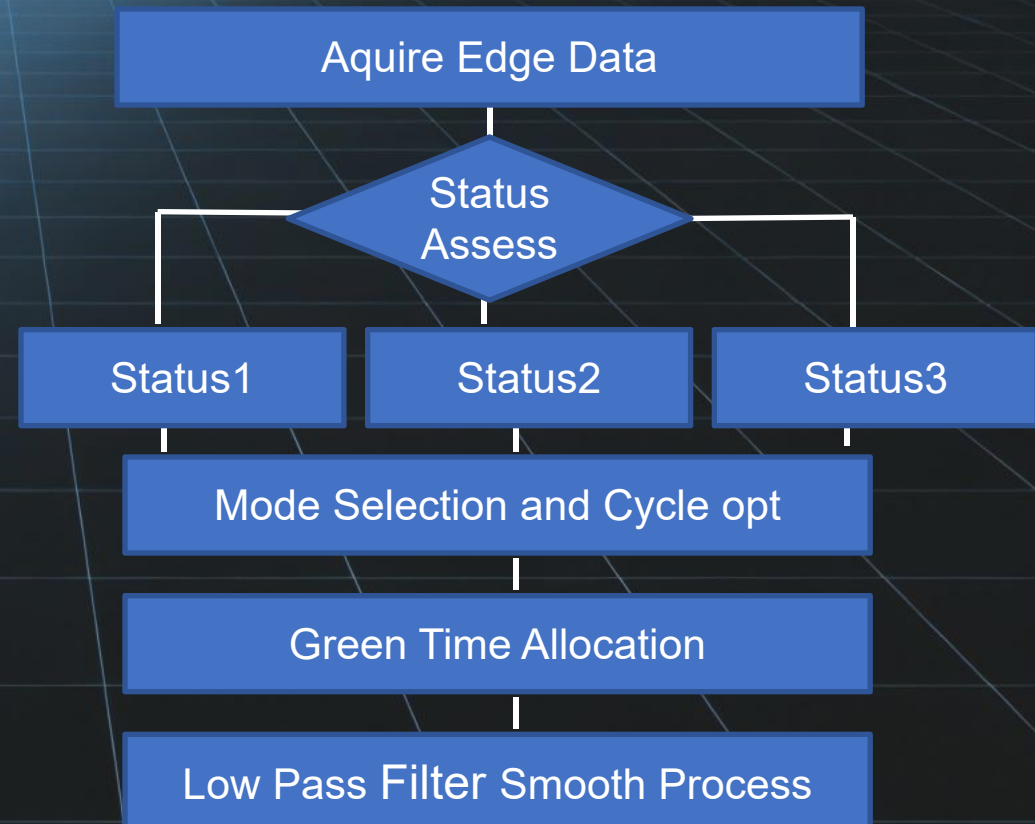


Adaptive Signal Control



Traffic Evaluation & Refinement

System Flowchart



Front-end Perception



Temporal Statistical Data

- Traffic flow monitoring
- Time occupancy analysis
- Vehicle headway measurement



Spatial Statistical Data

- Queue length detection
- Zone vehicle count
- Spatial occupancy rate



Core Capability: Accurately detects road traffic flow information by acquiring multi-dimensional temporal and spatial data.

Signal Control



Timing Prediction

Acquires temporal statistics to predict optimal timing plans, laying the foundation for efficient traffic flow.



Real-time Control

Uses real-time spatial data for precise phase switching, responding instantly to current traffic conditions.

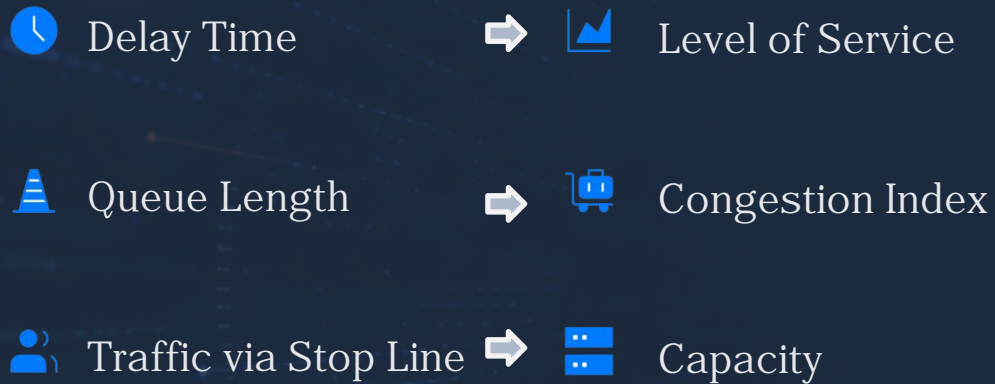


Adaptive Adjustment

Automatically adjusts phase split online based on detector data, enabling fully adaptive control strategies.

Traffic Evaluation & Refinement

Evaluation Indicators



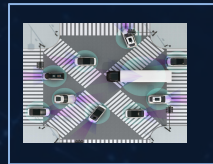
Core Capability

The system evaluates optimization effectiveness by comparing traffic parameters before and after implementation. This comprehensive analysis ensures that we can quantify the impact of our strategies and drive continuous improvement in traffic management.

Multi-Dimensional Effectiveness Evaluation



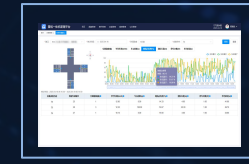
Safety Dimension



Efficiency Dimension



Equity Dimension



System Performance
Dimension

Multi-source Data Fusion

- Phase, Stage, and Cycle Data Collection
- Automatic Timing Plan Validation

Signal Controller Integration

Timing Plan Collection

Captures phase, stage, and cycle data from signal controller timing plans

Timing Plan Validation

Auto-validates plan accuracy using real-time traffic flow data

- All-Weather Capability (Rain/Snow/Night)
- Comprehensive Perception of People, Vehicles, and Roads

Radar-Video Fusion



- Vehicle Detection: Lane-level precise positioning with multi-type classification
- Vehicle Tracking: Trajectory tracking for accurate flow calculation in complex lanes
- Vehicle Classification: Distinguishes cars, vans, buses, trucks, and more

- Comprehensive: Full-Space Modeling.
- Granular: Lane-Level and Signal-Cycle-Level Metrics
- Accurate: 95%+ Accuracy for Traffic Flow and Key Metrics

Indicator Calculation

	#	Indicator Name
	1	Traffic Volume
	2	Average vehicle speed
	3	Congestion Evaluation
	4	Queue length
	5	Time headway
	6	Space headway
	7	Space occupancy rate
	8	Time occupancy rate
	9	Queue Dissipation Time
	10	Saturation Flow Rate
	11	Lost Time
	12	Queue Length at Green Onset
	13	Saturation Degree

Typical Intelligent Traffic Light Scenarios



Inductive Control Mode

For small to medium urban intersections, detects vehicle presence to adjust signal timing efficiently.



Adaptive Control Mode

For large urban intersections, uses real-time radar-video data to generate dynamic timing plans, solving queuing issues.



Dynamic Green Wave Mode

For urban arterial roads, maximizes green wave bandwidth based on real-time data to enhance corridor efficiency.

E2E Safety Assurance

Hardware Security



- **Hard Yellow Flash:** Dedicated board for critical exceptions.
- **Surge Protection:** Robust defense against lightning/strikes.
- **Hardware Redundancy:** Backup critical circuit boards.
- **Backup Power:** Continuous operation during outages.
- **Leakage Protection:** Built-in hazard detection.

Software & Logic Security



- **Soft Yellow Flash:** Software-triggered on logical errors.
- **Config Self-Check:** Automatic integrity verification.
- **Proactive Prevention:** Logic safeguards against errors.
- **Safe Startup:** Secure initialization to avoid unsafe states.

System-Level Security



- **Environmental Monitoring:** Tracks temp/humidity for reliability.
- **Degradation & Fault-Tolerant Operation:** Auto-detection of module failures (detectors, coordination, etc.), Smooth degradation to safer preset modes (E.G, Coordinated → Single-point adaptive etc), Prevents intersection disruption.

Timing Plan Management



Plan Access & Modification

Automatic Access and Update of Timing Plans
Access of Full Scheduling Plans and Time-of-Day
Tables



Diagnosis & Timing Recommendations

Our system analyzes plans and provides intelligent
recommendations for optimal timing adjustments.



Manual Secondary Editing

Fine-tune timing plans with a user-friendly interface
for manual adjustments and customization.



Historical Plan Query

Effortlessly retrieve and compare historical timing
plans to track changes and performance over time.

03

Pilot Effect Evaluation

Pilot Project Effect Evaluation

Indicator Overview and Analysis

Core KPIs (Volume, Speed, Delay Index) with MoM/WoW comparison — intuitively showing intersection performance trends.

Key Indicator Ranking Analysis

Ranked by Volume, Queue Length, and Avg Delay — Quickly pinpoint congestion hotspots.

Space Resource Comparison

Saturation vs. Road Space (Lanes, Width) — Evaluate efficiency, identify mismatches.

Trend and Pattern Analysis

Charts for daily/weekly/monthly trends — Cyclical patterns & anomaly detection

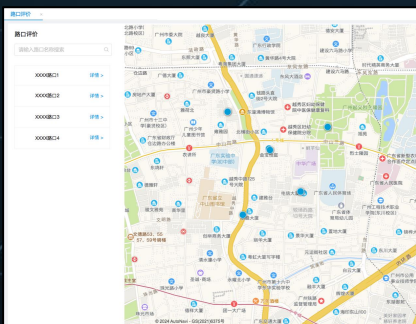
Time-Period Characteristic Evaluation

- Peak, off-peak, low-peak analysis
- Pinpoint bottleneck periods
- Inform differentiated signal timing

One-Click Export of Multi-Dimensional Data

- Export in multiple formats
- Support analysis, reporting, archiving

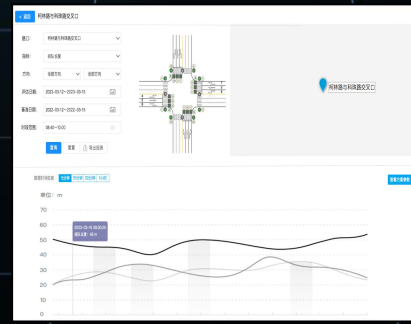
(PoP) Indicator Analysis



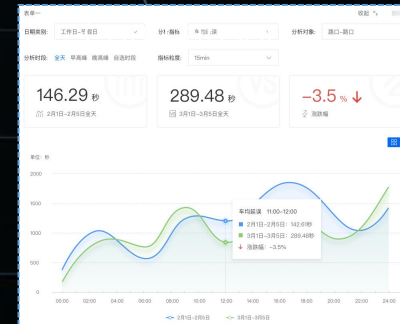
Space Resource Comparison



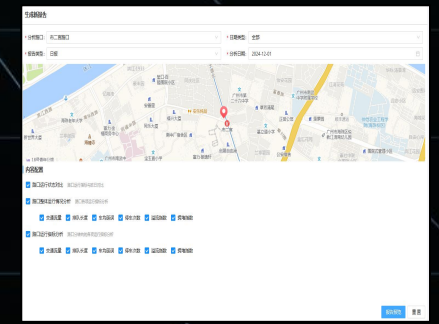
Trend Comparison



Time-Period



Evaluation Report



04

Implementation and Deployment Plan

Implementation and Deployment Plan

Phase 1 (W1-2): Preparation

Site Survey, Plan Confirmation,
Equipment Preparation

Phase 2 (W3): Installation

Installation, cabling, and
power-up of all hardware
devices

Phase 3 (W4-5): Commissioning

Radar calibration, lane
mapping, and system
integration testing.

Phase 4 (W6-8): Data Collection

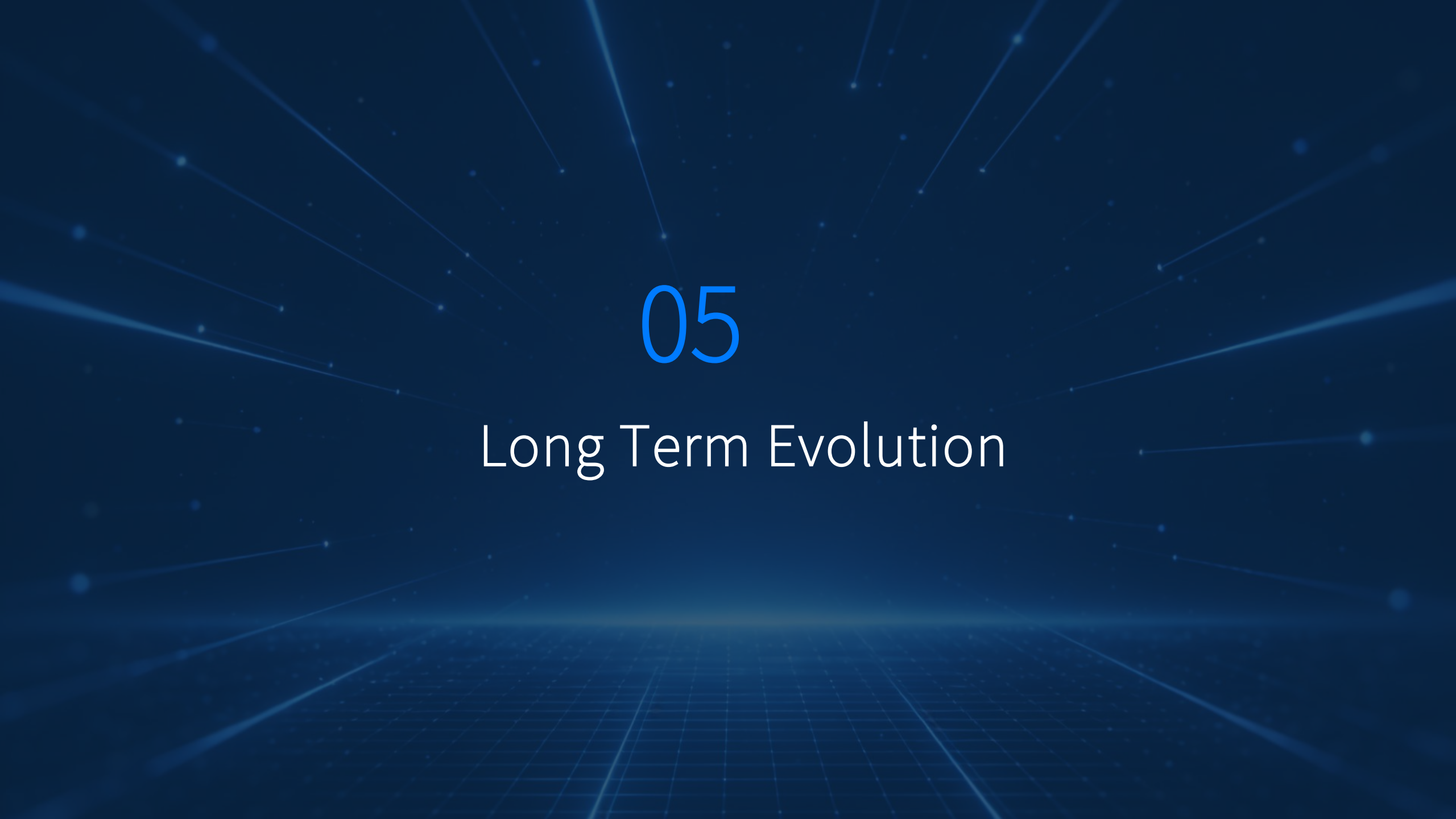
Operate in traditional fixed-
time mode to collect baseline
data for comparison.

Phase 5 (W9-10): Validation

Switch to adaptive mode to
collect optimized data.

Phase 6 (W11): Conclusion

Conduct comprehensive data
comparison and analysis, and
output report.

The background is a deep blue gradient. A perspective grid of thin, light blue lines extends from the bottom towards the center, creating a sense of depth. Numerous thin, light blue lines radiate outwards from the center, some ending in small, bright blue dots. The overall effect is futuristic and technological.

05

Long Term Evolution

Future Evolution Path



Intersection Enhancement

Introduce cutting-edge technologies such as Vehicle-to-Everything (V2X) to enhance single-point perception and control capabilities.



Intersection Coordination

Single-point to multi-intersection coordinated control enabling regional-level traffic coordination and management.



Integrated O&M

Integrate with existing signal controller networks to achieve unified device access and efficient operations and maintenance management.



Cross-Department Collaboration

Break down data silos to enable data sharing and interact with public security, special services, and other departments.

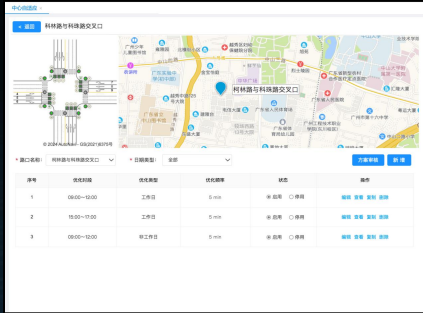


Towards a City Brain

Build a globally optimal intelligent traffic decision-making center based on full-scale data and AI algorithms.

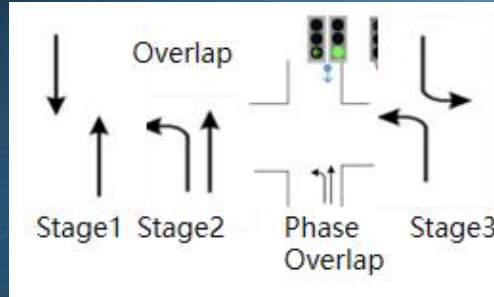
Single → Multi-Intersection Coordination EnerTraffic

Single intersection enhancement



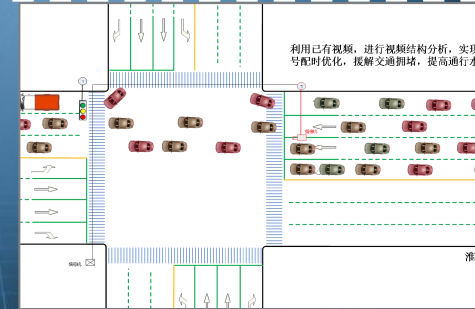
Single-Point Enhancement

Fine-tune single-point timing plans based on real-time multi-dimensional information.



Phase Overlap

Flexibly adjust phase overlap while maintaining the phase sequence within the ring.



Overflow Prevention & Control

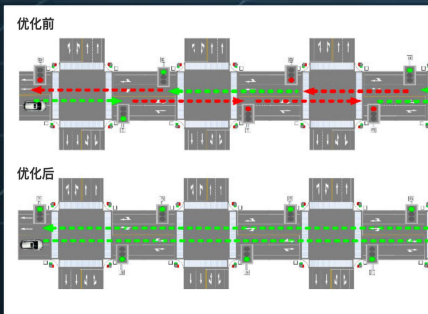
Monitor overflow risks in real time and implement relief measures at bottleneck intersections to prevent traffic gridlock.



Variable Lane Control

Adjust variable lane directions in real time to improve intersection efficiency from a spatial perspective.

Multi-Intersection Coordination&Optimization



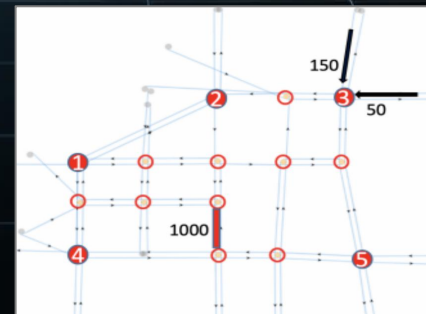
Arterial Coordination Control

Adjust intersection offsets to enable green wave passage and improve arterial traffic efficiency.



Green Wave Optimization

Speed-based offset fine-tuning with periodic split adjustment for smooth, stable progression.



Arterial Metering Control

Upstream control during peak hours to protect oversaturated areas and reduce central city congestion.

Regional Traffic Optimization



Regional Traffic Demand Mng.

Boundary control nodes regulate entry based on real-time congestion during peak hours.

Unified Signal Control

Integrated Signal
Network Control

Four Unifications

Data Fusion &
Applications

Unified Business Mng.

- Unified interface for all operations
- Cross-brand special service coordination
- Cross-brand coordinated control

Unified Signal Mng.

- Interconnection via standard interfaces for unified management.

Unified Interface Mng.

- Integrate with existing traffic systems
- Enable Vehicle-Road-Cloud integrated applications

Unified Data Services

- Multi-source data fusion
- Standardized data sharing mechanism

Signal Timing and
Status Push

Integration with
Superior Platform

Indicator Data Collection

Multi-Source Data Fusion

Signal Optimization
Effectiveness Evaluation

Multi-Dimensional
Statistical Analysis

Networking & Integrated
Control

Point-Line-Area
Coordinated Optimization

Public Transport
Priority

Special Service Assurance

Multi-Brand Signal
Controller Compatibility

Signal Status and
Plan Monitoring

Intersection Inventory

Signal Controller O&M

Cross-Department Collaboration Toward the City Brain



Special Event Security

- Real-time data + intelligent signal control
- Dynamic route planning for VIPs & major events
- Ensure safe and smooth passage



Emergency Management

- Rapid emergency vehicle identification
- Signal priority control
- Open a "green corridor" for critical response



Municipal Urban Management

- Traffic impact assessment for road works
- Route optimization for sanitation vehicles
- Nighttime construction noise monitoring & control



Traffic Enforcement

Utilizes high-definition video and AI recognition technology to automatically capture various traffic violations (such as running red lights, illegal parking, etc.), improving enforcement efficiency and fairness.



THANKS

Should you need further information or have any questions about this proposal, please feel free to contact us.

EnerTraffic

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